US Patent & Trademark Office
Patent Public Search | Text View

United States Patent Application Publication	20250287089
Kind Code	A1
Publication Date	September 11, 2025
Inventor(s)	Zanoff; Michael C.

ATTACHMENT FOR BORE INSPECTION SYSTEM

Abstract

An attachment for a bore inspection system for inspecting a bore of a component includes a first section adapted to be removably coupled to an imaging device of the bore inspection system. The first section extends along a central axis. The attachment also includes a fastener adapted to couple the first section with the imaging device and a second section integral with the first section and extending from the first section along the central axis. The second section includes a mirror that is disposed at an inclination angle of 45 degrees relative to the central axis. When the attachment is coupled to the imaging device, the mirror faces a lens of the imaging device so that light rays reflected from a bore wall of the bore, and to be received by the imaging device, are bent by an angle of 90 degrees by the mirror to inspect the bore wall.

Inventors: **Zanoff; Michael C. (Westworth Village, TX)**

Applicant: **SPM Oil & Gas Inc.** (Fort Worth, TX)

Family ID: **1000007738537**

Assignee: **SPM Oil & Gas Inc. (Fort Worth, TX)**

Appl. No.: **18/595897**

Filed: **March 05, 2024**

Publication Classification

Int. Cl.: **H04N23/50** (20230101); **F16L55/40** (20060101); **F16L101/30** (20060101); **H04N23/55** (20230101)

U.S. Cl.:

CPC **H04N23/555** (20230101); **F16L55/40** (20130101); **H04N23/55** (20230101); F16L2101/30 (20130101)

Background/Summary

TECHNICAL FIELD

[0001] The present disclosure relates to a bore inspection system, and more particularly, to an attachment for the bore inspection system.

BACKGROUND

[0002] Bores of components, such as engines or fracking equipment, may have to be periodically inspected in order to determine any damage or failure of one or more components of such components. The bores of the components are also inspected after manufacturing/assembly to ensure compliance with design requirements. Typically, an imaging device is used to determine a current condition of a bore wall of such bores. The imaging device are received within the bore of the component to capture images or videos of the bore wall of the bore. In some examples, it may be desired to take close-up images/videos of the bore wall. However, some bores are too narrow or small in diameter for the imaging device to be turned sideways to inspect the bore wall. Hence, a solution is desired to capture images or videos of the bore wall.

[0003] U.S. Pat. No. 8,587,647 describes a remote inspection device imager assembly that includes an imager body having a male threaded portion. An accessory assembly includes: a tubular body portion having first internal female threads engaged with the male threaded portion such that tubular body portion rotation axially translates the tubular body portion with respect to the imager body; and a mirror obliquely angled with respect to a longitudinal axis of both the imager assembly. A threaded coupler positioned between the imager body and tubular body portion has second internal female threads engaged with the male threaded portion. The threaded coupler is selectively axially translated by rotation to a first contact position with the imager body or a second contact position with the tubular body portion. The second contact position binds the first and second internal female threads with the male threaded portion to prevent tubular body portion axial rotation and fix a mirror orientation.

SUMMARY OF THE DISCLOSURE

[0004] In an aspect of the present disclosure, an attachment for a bore inspection system for inspecting a bore of a component is provided. The attachment includes a first section adapted to be removably coupled to an imaging device of the bore inspection system. The first section extends along a central axis. The attachment also includes a fastener adapted to couple the first section with the imaging device. The attachment further includes a second section integral with the first section and extending from the first section along the central axis. The second section includes a mirror that is disposed at an inclination angle of 45 degrees relative to the central axis. Further, when the attachment is coupled to the imaging device, the mirror is adapted to face a lens of the imaging device so that light rays reflected from a bore wall of the bore, and to be received by the imaging device, are bent by an angle of 90 degrees by the mirror to inspect the bore wall.

[0005] In another aspect of the present disclosure, a bore inspection system for inspecting a bore of a component is provided. The bore inspection system includes an imaging device defining a first end and a second end. The imaging device includes a lens disposed proximal to the second end. The bore inspection system also includes an attachment. The attachment includes a first section adapted to be removably coupled to the imaging device of the bore inspection system. The first section extends along a central axis. The attachment also includes a fastener adapted to couple the first section with the imaging device. The attachment further includes a second section integral with the first section and extending from the first section along the central axis. The second section includes a mirror that is disposed at an inclination angle of 45 degrees relative to the central axis. Further, when the attachment is coupled to the imaging device, the mirror is adapted to face the lens of the imaging device so that light rays reflected from a bore wall of the bore, and to be received by the imaging device, are bent by an angle of 90 degrees by the mirror to inspect the bore wall.

[0006] In yet another aspect of the present disclosure, a method for inspecting a bore of a fracturing equipment is provided. The method includes providing an imaging device of a bore inspection system. The imaging device defines a first end and a second end. The imaging device includes a lens disposed proximal to the second end. The method also includes providing an attachment of the bore inspection system. The attachment includes a first section extending along a central axis, a fastener, and a second section integral with the first section and extending from the first section along the central axis. The second section includes a mirror that is disposed at an inclination angle of 45 degrees relative to the central axis. The method further includes coupling, via the fastener of the attachment, the first section of the attachment with the imaging device, such that the mirror of the second section faces the lens of the imaging device. The method includes inserting the bore inspection system within the bore of the fracturing equipment. The method also includes bending, by the mirror of the second section, light rays reflected from a bore wall of the bore, and receivable by the imaging device, by an angle of 90 degrees to inspect the bore wall.

[0007] Other features and aspects of this disclosure will be apparent from the following description and the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIG. **1** illustrates an exemplary component and a bore inspection system for inspecting a bore of the component, according to an embodiment of the present disclosure;

[0009] FIG. **2** is a schematic perspective view of the bore inspection system of FIG. **1**;

[0010] FIG. **3** is a schematic perspective view of an attachment of the bore inspection system of FIG. **2**, according to an embodiment of the present disclosure;

[0011] FIG. **4** is a schematic cross-sectional side view illustrating the bore inspection system of FIG. **2** inserted within the bore of the component; and

[0012] FIG. **5** is a flowchart for a method of inspecting the bore of a fracturing equipment, according to an embodiment of the present disclosure.

DETAILED DESCRIPTION

[0013] Wherever possible, the same reference numbers will be used throughout the drawings to refer to the same or like parts.

[0014] Referring to FIG. **1**, an exemplary component **100** is illustrated. The component **100** includes, and may be interchangeably referred to as, a fracturing equipment **100** herein. Specifically, the fracturing equipment **100** includes a fluid end of a fracturing pump. Alternatively, the component **100** may include an internal combustion engine, a gas turbine engine, a pipe, a tube, or any other component that has bores, holes, openings, and the like. The component **100** includes a bore **102**. The bore **102** defines a bore wall **104**.

[0015] FIG. **1** also illustrates a bore inspection system **200** for inspecting the bore **102** of the component **100**. It should be noted that the bore inspection system **200** may be used to inspect bores, holes, or openings of different shapes, such as, a circular shape, a square shape, a rectangular shape, an elliptical shape, a triangular shape, and the like. As illustrated in FIG. **1**, the bore inspection system **200** is inserted into the bore **102** of the component **100** for inspection. The bore inspection system **200** may be used to inspect the bore wall **104** to detect damage or failures in the bore wall **104**. Further, the bore inspection system **200** may be used to inspect the bore wall **104** to check compliance with desired design requirements. In some examples, the bore inspection system **200** may be used to inspect components or parts disposed in confined spaces or in machineries that may not be feasible to disassemble for minor inspection/repair purposes.

[0016] Referring to FIG. **2**, the bore inspection system **200** includes an imaging device **202**. The imaging device **202** may include a microscopic handheld camera, for example. It should be noted that the imaging device **202** may include any type of device that can capture images or videos. The imaging device **202** defines a first end **204** and a second end **206**. The imaging device **202** includes a housing **208**. The housing **208** is cylindrical in shape. The housing **208** has an outer profile **210** that is circular in shape. However, the outer profile **210** may be rectangular in shape, or square in shape, or elliptical in shape, as per application requirements. The imaging device **202** also includes a lens **212** disposed proximal to the second end **206**. The lens **212** is disposed within the housing **208**. The lens **212** may include an objective lens that gathers light rays from an object being observed and focuses the light rays to produce an image or video of the object.

[0017] The present disclosure relates to an attachment **300** for the bore inspection system **200** for inspecting the bore **102** (see FIGS. **1** and **4**) of the component **100** (see FIGS. **1** and **4**). Referring to FIGS. **2** and **3**, the bore inspection system **200** includes the attachment **300**. The attachment **300** may be made from a metal, an alloy, or a polymer. The attachment **300** includes a first section **302** that is removably coupled to the imaging device **202** of the bore inspection system **200**. The first section **302** extends along a central axis X**1**. The attachment **300** also includes a fastener **304** adapted to couple the first section **302** with the imaging device **202**. The attachment **300** further includes a second section **306** integral with the first section **302** and extending from the first section **302** along the central axis X**1**.

[0018] Referring now to FIG. **3**, the first section **302** includes a coupling portion **308**. The coupling portion **308** defines a through-opening **310** to receive a portion of the imaging device **202** (see FIG. **2**) therein. The through-opening **310** has a shape that corresponds to the outer profile **210** (see FIG. **2**) of the portion of the imaging device **202**. In the illustrated example of FIG. **3**, the through-opening **310** has a circular shape. Specifically, the circular shape of the through-opening **310** allows the imaging device **202** to be partially received within the coupling portion **308**. Alternatively, the through-opening **310** may have a rectangular shape, a square shape, an elliptical shape, based on the shape of the outer profile **210**. It should be noted that a dimension of the coupling portion **308** may vary based on dimensions of the imaging device **202**.

[0019] Further, the first section **302** includes a first projection **312** and a second projection **314**. Each of the first projection **312** and the second projection **314** extends from the coupling portion **308** along a first direction F**1** that is orthogonal to the central axis X**1**. The first projection **312** and the second projection **314** are similar in shape and dimensions. Each of the first and second projections **312**, **314** are rectangular in shape. It should be noted that the first and second projections **312**, **314** may have any other shape as per application requirements. The coupling portion **308** is integral with each of the first projection **312** and the second projection **314**. Each of the first projection **312** and the second projection **314** is adapted to receive the fastener **304** to couple the first section **302** with the imaging device **202**. Specifically, the first projection **312** defines a first through-hole (not shown) and the second projection **314** defines a second through-hole (not shown). The first through-hole aligns with the second through-hole to receive the fastener **304** to couple the first section **302** with the imaging device **202**.

[0020] Further, the second section **306** includes an arcuate portion **320**, a base surface **322** extending from the arcuate portion **320** along the first direction F**1**, and a pair of side plates **324** coupled to each of the arcuate portion **320** and the base surface **322**. The base surface **322** and the pair of side plates **324** are integral with the arcuate portion **320**. The side plates **324** extend orthogonally from the base surface **322** along the central axis X**1**.

[0021] Moreover, the second section **306** includes a mirror **326** that is disposed at an inclination angle A**1** of 45 degrees relative to the central axis X**1**. The mirror **326** is retained within the second section **306** by each of the arcuate portion **320**, the base surface **322**, and the pair of side plates **324**. In some examples, the mirror **326** may be retained within the second section **306** using suitable coupling means, such as, adhesives, fasteners, and the like. Further, when the attachment **300** is coupled to the imaging device **202**, the mirror **326** faces the lens **212** of the imaging device **202** so that light rays reflected from the bore wall **104** (see FIG. **4**) of the bore **102** (see FIG. **4**), and to be received by the imaging device **202**, are bent by an angle of 90 degrees by the mirror **326** to inspect the bore wall **104**.

[0022] As shown in FIG. **3**, an outer edge **328** of the base surface **322** is in-line with an outer edge **330** of each of the first projection **312** and the second projection **314**. In other words, a distance D**1** between the central axis X**1** and the outer edge **328** of the base surface **322** is same as a distance D**2** between the central axis X**1** and the outer edge **330** of each of the first and second projections **312**, **314**. Further, the distance D**1** between the central axis X**1** and the outer edge **328** of the base surface **322**, and the distance D**2** between the central axis X**1** and the outer edge **330** of each of the first and second projections **312**, **314** is based on a desired stand-off distance D**3** (shown in FIG. **4**) defined between a central point P**1** of the mirror **326** and the bore wall **104**. It should be noted that the stand-off distance D**3** is decided based on a desired focal length of the imaging device **202**. Thus, a length of the base surface **322** and a length of each of the first and second projections **312**, **314** may be varied to achieve the desired stand-off distance D**3**.

[0023] Referring now to FIG. **4**, during an inspection of the bore wall **104**, the bore inspection system **200** is inserted into the bore **102**. Further, the central axis X**1** along which the attachment **300** extends is parallel to an axis X**2** of the bore **102**. Furthermore, the outer edge **328** of the base surface **322** and the outer edge **330** of each of the first projection **312** and the second projection **314** engage with the bore wall **104** to dispose the mirror **326** (see FIG. **3**) of the second section **306** at an angle of 45 degrees relative to the bore wall **104**. In other words, the attachment **300** is pressed against the bore wall **104** to ensure that the imaging device **202** is focused and perpendicular to the bore wall **104**. Specifically, when the attachment **300** is pressed against the bore wall **104**, such that the imaging device **202** is disposed at the desired stand-off distance D**3** from the bore wall **104**, which may improve a focus of the imaging device **202**.

[0024] It is to be understood that individual features shown or described for one embodiment may be combined with individual features shown or described for another embodiment. The above described implementation does not in any way limit the scope of the present disclosure. Therefore, it is to be understood although some features are shown or described to illustrate the use of the present disclosure in the context of functional segments, such features may be omitted from the scope of the present disclosure without departing from the spirit of the present disclosure as defined in the appended claims.

INDUSTRIAL APPLICABILITY

[0025] The present disclosure describes the attachment **300** for the bore inspection system **200**. The attachment **300** can be removably coupled to any type of image capturing device to inspect bores, holes, or openings in various components, such as, engines, tubes, pipes, fracking equipment such as pumps, and the like. In an example, the bore inspection system **200** including the attachment **300** may be particularly usable in inspecting bores having a bore size between 2 inches and 8 inches, for example, because in such bores it may be difficult to turn the imaging device **202** sideways for inspecting the bores. Further, the attachment **300** may be easily and quickly coupled with the imaging device **202** without modifying a design of the imaging device **202**. The attachment **300** may be easy to manufacture and provides a cost-effective solution to inspect the bore wall **104** of the bore **102**.

[0026] The attachment **300** includes the first projection **312**, the second projection **314**, and the base surface **322** that may be pressed against the bore wall **104** to ensure the bore wall **104** is in focus. Further, the distance D**1** between the central axis A**1** and the base surface **322**, and the distance D**2** between the central axis A**1** and each of the first and second projections **312**, **314** is decided based on the desired stand-off distance D**3**. The distances D**1**, D**2** may be varied by varying the dimensions of the base surface **322**, the first projection **312**, and the second projection **314** to achieve the desired stand-off distance D**3**. Further, when the imaging device **202** is disposed

at the desired stand-off distance D**3**, the imaging device **202** may generate images of optimum quality, which may improve a usability and a reliability of the bore inspection system **200**.
[0027] FIG. **5** is a flowchart for a method **500** for inspecting the bore **102** of the fracturing equipment **100**. Referring to FIGS. **1** to **5**, at step **502**, the imaging device **202** of the bore inspection system **200** is provided. The imaging device **202** defines the first end **204** and the second end **206**. The imaging device **202** includes the lens **212** disposed proximal to the second end **206**.
[0028] At step **504**, the attachment **300** of the bore inspection system **200** is provided. The attachment **300** includes the first section **302** extending along the central axis X**1**, the fastener **304**, and the second section **306** integral with the first section **302** and extending from the first section **302** along the central axis X**1**. The second section **306** includes the mirror **326** that is disposed at the inclination angle A**1** of 45 degrees relative to the central axis X**1**.
[0029] At step **506**, the fastener **304** of the attachment **300** couples the first section **302** of the attachment **300** with the imaging device **202**, such that the mirror **326** of the second section **306** faces the lens **212** of the imaging device **202**.
[0030] At step **508**, the bore inspection system **200** is inserted within the bore **102** of the fracturing equipment **100**. At the step **508**, the portion of each of the first section **302** and the second section **306** of the attachment **300** is engaged with the bore wall **104** of the bore **102** to dispose the mirror **326** of the second section **306** at the angle of 45 degrees relative to the bore wall **104** of the bore **102**.
[0031] At step **510**, the mirror **326** of the second section **306** bends light rays reflected from the bore wall **104** of the bore **102**, and receivable by the imaging device **202**, by the angle of 90 degrees to inspect the bore wall **104**.
[0032] While aspects of the present disclosure have been particularly shown and described with reference to the embodiments above, it will be understood by those skilled in the art that various additional embodiments may be contemplated by the modification of the disclosed work machine, systems and methods without departing from the spirit and scope of the disclosure. Such embodiments should be understood to fall within the scope of the present disclosure as determined based upon the claims and any equivalents thereof.

Claims

1. An attachment for a bore inspection system for inspecting a bore of a component, the attachment comprising: a first section adapted to be removably coupled to an imaging device of the bore inspection system, the first section extending along a central axis; a fastener adapted to couple the first section with the imaging device; and a second section integral with the first section and extending from the first section along the central axis, wherein, the second section includes a mirror that is disposed at an inclination angle of 45 degrees relative to the central axis, and when the attachment is coupled to the imaging device, the mirror is adapted to face a lens of the imaging device so that light rays reflected from a bore wall of the bore, and to be received by the imaging device, are bent by an angle of 90 degrees by the mirror to inspect the bore wall.
2. The attachment of claim 1, wherein the first section includes a coupling portion, wherein the coupling portion defines a through-opening adapted to receive a portion of the imaging device therein, and wherein the through-opening has a shape that corresponds to an outer profile of the portion of the imaging device.
3. The attachment of claim 2, wherein the through-opening has a circular shape.
4. The attachment of claim 2, wherein the first section includes a first projection and a second projection, wherein each of the first projection and the second projection extends from the coupling portion along a first direction that is orthogonal to the central axis, and wherein each of the first projection and the second projection is adapted to receive the fastener to couple the first section with the imaging device.

5. The attachment of claim 4, wherein the second section includes an arcuate portion, a base surface extending from the arcuate portion along the first direction, and a pair of side plates coupled to each of the arcuate portion and the base surface.

6. The attachment of claim 5, wherein an outer edge of the base surface is in-line with an outer edge of each of the first projection and the second projection, and wherein, during an inspection of the bore wall, the outer edge of the base surface and the outer edge of each of the first projection and the second projection is adapted to engage with the bore wall to dispose the mirror of the second section at an angle of 45 degrees relative to the bore wall.

7. The attachment of claim 6, wherein a distance between the central axis and the outer edge of the base surface, and a distance between the central axis and the outer edge of each of the first and second projections is based on a desired stand-off distance defined between a central point of the mirror and the bore wall.

8. The attachment of claim 1, wherein the component includes a fracturing equipment.

9. The attachment of claim 8, wherein the fracturing equipment includes a fluid end of a fracturing pump.

10. A bore inspection system for inspecting a bore of a component, the bore inspection system comprising: an imaging device defining a first end and a second end, wherein the imaging device includes a lens disposed proximal to the second end; and an attachment including: a first section adapted to be removably coupled to the imaging device of the bore inspection system, the first section extending along a central axis; a fastener adapted to couple the first section with the imaging device; and a second section integral with the first section and extending from the first section along the central axis, wherein, the second section includes a mirror that is disposed at an inclination angle of 45 degrees relative to the central axis, and when the attachment is coupled to the imaging device, the mirror is adapted to face the lens of the imaging device so that light rays reflected from a bore wall of the bore, and to be received by the imaging device, are bent by an angle of 90 degrees by the mirror to inspect the bore wall.

11. The bore inspection system of claim 10, wherein the first section includes a coupling portion, wherein the coupling portion defines a through-opening adapted to receive a portion of the imaging device therein, and wherein the through-opening has a shape that corresponds to an outer profile of the portion of the imaging device.

12. The bore inspection system of claim 11, wherein the through-opening has a circular shape.

13. The bore inspection system of claim 11, wherein the first section includes a first projection and a second projection, wherein each of the first projection and the second projection extends from the coupling portion along a first direction that is orthogonal to the central axis, and wherein each of the first projection and the second projection is adapted to receive the fastener to couple the first section with the imaging device.

14. The bore inspection system of claim 13, wherein the second section includes an arcuate portion, a base surface extending from the arcuate portion along the first direction, and a pair of side plates coupled to each of the arcuate portion and the base surface.

15. The bore inspection system of claim 14, wherein an outer edge of the base surface is in-line with an outer edge of each of the first projection and the second projection, and wherein, during an inspection of the bore wall, the outer edge of the base surface and the outer edge of each of the first projection and the second projection is adapted to engage with the bore wall to dispose the mirror of the second section at an angle of 45 degrees relative to the bore wall.

16. The bore inspection system of claim 15, wherein a distance between the central axis and the outer edge of the base surface, and a distance between the central axis and the outer edge of each of the first and second projections is based on a desired stand-off distance defined between a central point of the mirror and the bore wall.

17. The bore inspection system of claim 10, wherein the component includes a fracturing equipment.

18. The bore inspection system of claim 17, wherein the fracturing equipment includes a fluid end of a fracturing pump.

19. A method for inspecting a bore of a fracturing equipment, the method comprising: providing an imaging device of a bore inspection system, the imaging device defining a first end and a second end, wherein the imaging device includes a lens disposed proximal to the second end; providing an attachment of the bore inspection system, wherein the attachment includes a first section extending along a central axis, a fastener, and a second section integral with the first section and extending from the first section along the central axis, and wherein the second section includes a mirror that is disposed at an inclination angle of 45 degrees relative to the central axis; coupling, via the fastener of the attachment, the first section of the attachment with the imaging device, such that the mirror of the second section faces the lens of the imaging device; inserting the bore inspection system within the bore of the fracturing equipment; and bending, by the mirror of the second section, light rays reflected from a bore wall of the bore, and receivable by the imaging device, by an angle of 90 degrees to inspect the bore wall.

20. The method of claim 19, wherein the step of inserting the bore inspection system within the bore of the fracturing equipment further includes engaging a portion of each of the first section and the second section of the attachment with the bore wall to dispose the mirror of the second section at an angle of 45 degrees relative to the bore wall.
